config.py

`src/moe_pruning/config.py · 405 lines · module moe_pruning.config`

What it does

Loads, validates and exposes every setting of the project as one frozen `ExperimentConfig` object. All scripts and library code read settings from this object; nothing reads environment variables directly.

Settings are layered: `.env` (lowest), then model profiles such as `configs/olmoe-1b-7b-0125.env` passed with `--profile`, then real environment variables, then `--set KEY=VALUE` overrides (highest).

How it works

- `load_config()` collects the raw values, converts them with typed readers (string, choice, int, float, path) and collects every problem before raising a single `ConfigError`, so all mistakes are reported at once.
- Relative paths are resolved against the project root, so scripts work from any working directory.
- Extra checks: a non-empty `MODEL_PATH` must be an existing directory with a `config.json` (no silent fallback to a download), C4 length bounds must be ordered, `DEVICE_MAP` / `MAX_MEMORY` must parse.
- Derived locations: `run_dir`, `run_stats_dir` and `run_cache_dir` put every artifact of one experiment under `EXPERIMENT_NAME`; `model_source` is the local path if set, otherwise the Hugging Face id.
- `add_config_arguments()` / `config_from_args()` add `--env-file`, `--profile` and `--set` to every script.

Notes

- Valid choices are defined at the top of the file (`VALID_DTYPES`, `VALID_REPLACEMENT_METHODS`, ...); adding a new option starts here.
- `to_dict()` is what gets written into every config snapshot and manifest.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class ConfigError`	Raised when the configuration is missing or invalid.
`class ExperimentConfig`	Frozen, validated settings of one run (one field per .env variable).
`.run_dir() [property]`	Directory holding every artifact of the current experiment.
`.run_stats_dir() [property]`	Statistics directory of this experiment.
`.run_cache_dir() [property]`	Cache directory of this experiment.
`.model_source() [property]`	Local path if configured, otherwise the Hugging Face model id.
`.to_dict()`	Convert to a plain dictionary for JSON reports.
`load_config(env_file, overrides, profiles)`	Load and validate the configuration.
`add_config_arguments(parser)`	Add --env-file, --profile and --set to a script's argument parser.
`config_from_args(args)`	Build the configuration from parsed command-line arguments.

Settings it reads

`C4_MAX_TEXT_LENGTH`, `C4_MIN_TEXT_LENGTH`, `EXPERIMENT_NAME`

Connections

Uses: `moe_pruning.model.placement`

Used by: `src/moe_pruning/evaluation/evaluator.py`, `src/moe_pruning/model/loader.py`, `src/moe_pruning/pruning/manifest.py`, `src/moe_pruning/utils/cli.py`, `src/moe_pruning/utils/paths.py`